

## 종합실습 1 : 비동기 ETL 파이프라인

### step 0~2 : Transform 부터 만든다 / 테스트를 바로 붙인다

#### 실행 결과

```
(.venv) gayeon@sunggayeonui-MacBookPro skala_python % pytest -v Python_data_0720_Day1/Total1/test_pipeline.py
===== test session starts =====
platform darwin -- Python 3.14.6, pytest-9.1.1, pluggy-1.6.0 -- /Users/gayeon/skala_python/.venv/bin/python3.14
cachedir: .pytest_cache
rootdir: /Users/gayeon/skala_python
plugins: asyncio-1.4.0, anyio-4.14.2
asyncio: mode=Mode.STRICT, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function
collected 3 items

Python_data_0720_Day1/Total1/test_pipeline.py::test_category_lowercase PASSED [ 33%]
Python_data_0720_Day1/Total1/test_pipeline.py::test_negative_price_rejected PASSED [ 66%]
Python_data_0720_Day1/Total1/test_pipeline.py::test_valid_invalid_count PASSED [100%]

===== 3 passed in 0.05s =====
(.venv) gayeon@sunggayeonui-MacBookPro skala_python %
```

### STEP 3~6: 비동기 수집(Extract) → Pydantic 검증(Transform) → CSV·Parquet 저장(Load)

#### → run() 호출

#### 실행 결과

```
(.venv) gayeon@sunggayeonui-MacBookPro skala_python %
=====
비동기 ETL 파이프라인 실행 결과
=====
전체 수집 : 60건
유효 데이터 : 60건
오염 데이터 : 0건
저장 데이터 : 60건
(.venv) gayeon@sunggayeonui-MacBookPro skala_python %
```

- 비동기로 60건을 수집한 뒤 Pydantic으로 검증하여 60건 모두 정상 데이터로 분류
- 검증된 데이터는 products.csv와 products.parquet으로 저장

```
(.venv) gayeon@sunggayeonui-MacBookPro skala_python % pytest -v Python_data_0720_Day1/Total1/test_pipeline.py
===== test session starts =====
platform darwin -- Python 3.14.6, pytest-9.1.1, pluggy-1.6.0 -- /Users/gayeon/skala_python/.venv/bin/python3.14
cachedir: .pytest_cache
rootdir: /Users/gayeon/skala_python
plugins: asyncio-1.4.0, anyio-4.14.2
asyncio: mode=Mode.STRICT, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function
collected 6 items

Python_data_0720_Day1/Total1/test_pipeline.py::test_category_lowercase PASSED [ 16%]
Python_data_0720_Day1/Total1/test_pipeline.py::test_negative_price_rejected PASSED [ 33%]
Python_data_0720_Day1/Total1/test_pipeline.py::test_valid_invalid_count PASSED [ 50%]
Python_data_0720_Day1/Total1/test_pipeline.py::test_missing_field_rejected PASSED [ 66%]
Python_data_0720_Day1/Total1/test_pipeline.py::test_load_creates_files PASSED [ 83%]
Python_data_0720_Day1/Total1/test_pipeline.py::test_parquet_roundtrip PASSED [100%]

===== 6 passed in 3.49s =====
(.venv) gayeon@sunggayeonui-MacBookPro skala_python %
```

- 6개 테스트 모두 통과

## step 7 : total1 폴더 검사 (ruff)

### 실행 결과

```
===== 6 passed in 3.49s =====
(.venv) gayeon@sunggayeonui-MacBookPro skala_python % ruff check Python_data_0720_Day1/Total1
All checks passed!
(.venv) gayeon@sunggayeonui-MacBookPro skala_python % ruff check Python_data_0720_Day1/Total1 --fix
All checks passed!
(.venv) gayeon@sunggayeonui-MacBookPro skala_python % ruff format Python_data_0720_Day1/Total1
3 files reformatted
(.venv) gayeon@sunggayeonui-MacBookPro skala_python % ruff check Python_data_0720_Day1/Total1
All checks passed!
(.venv) gayeon@sunggayeonui-MacBookPro skala_python % pytest -v Python_data_0720_Day1/Total1/test_pipeline.py
===== test session starts =====
platform darwin -- Python 3.14.6, pytest-9.1.1, pluggy-1.6.0 -- /Users/gayeon/skala_python/.venv/bin/python3.14
cachedir: .pytest_cache
rootdir: /Users/gayeon/skala_python
plugins: asyncio-1.4.0, anyio-4.14.2
asyncio: mode=Mode.STRICT, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function
collected 6 items

Python_data_0720_Day1/Total1/test_pipeline.py::test_category_lowercase PASSED [ 16%]
Python_data_0720_Day1/Total1/test_pipeline.py::test_negative_price_rejected PASSED [ 33%]
Python_data_0720_Day1/Total1/test_pipeline.py::test_valid_invalid_count PASSED [ 50%]
Python_data_0720_Day1/Total1/test_pipeline.py::test_missing_field_rejected PASSED [ 66%]
Python_data_0720_Day1/Total1/test_pipeline.py::test_load_creates_files PASSED [ 83%]
Python_data_0720_Day1/Total1/test_pipeline.py::test_parquet_roundtrip PASSED [100%]

===== 6 passed in 0.48s =====
(.venv) gayeon@sunggayeonui-MacBookPro skala_python %
```

```
(.venv) gayeon@sunggayeonui-MacBookPro skala_python % python Python_data_0720_Day1/Total1/pipeline.py
=====
비동기 ETL 파이프라인 실행 결과
=====
전체 수집 : 60건
유효 데이터 : 60건
오염 데이터 : 0건
저장 데이터 : 60건
(.venv) gayeon@sunggayeonui-MacBookPro skala_python %
```

- Ruff 검사에서 오류가 발견되지 않았으며 코드 형식 정리 후에도 파이프라인과 테스트가 정상 실행됨을 확인